

MathsGen: Mixed GCSE Practice

Practice worksheet

Name: _____ Date: _____

Question 1 ★★★★★ [+ New](#) | [Answer](#) | [Flag](#) | [Options](#)

Simplify by collecting like terms · algebra.expressions.like_terms v1

Simplify

$$-3x + 8y + 2yx + 6xy - 8x$$

Question 2 ★★★ [+ New](#) | [Answer](#) | [Flag](#) | [Options](#)

Change the subject of a formula · algebra.rearranging.changing_subject v3

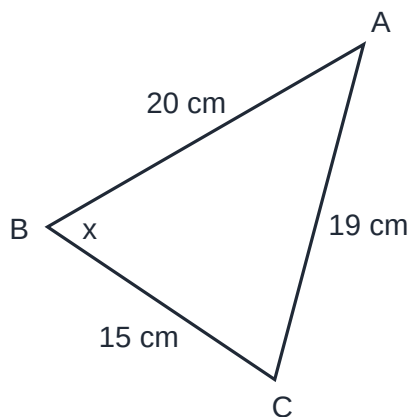
Make x the subject. All other letters represent positive numbers. The denominator in the formula is nonzero.

$$g = \frac{p}{8x + n}$$

Question 3 ★★ | [+ New](#) | [Answer](#) | [Flag](#) | [Options](#)

Cosine rule: sides, angles, area and linked triangles · geometry.trigonometry.cosine_rule v4

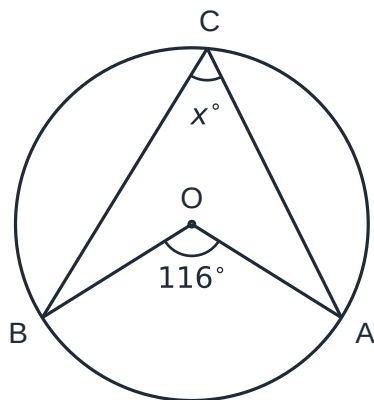
In triangle ABC, $AB = 20$ cm, $AC = 19$ cm and $BC = 15$ cm. Find angle ABC. Give answers to 1 decimal place.



Question 4 ★★ | [+ New](#) | [Answer](#) | [Flag](#) | [Options](#)

Circle theorem: angle at the centre · geometry.circle_theoremscentre_angle v2

O is the centre of the circle. A, B and C lie on the circumference. Find x .



Question 5 ★★★ [+ New](#) | [Answer](#) | [Flag](#) | [Options](#)

Venn diagrams: calculating probabilities · probability.venn.probabilities v1

The Venn diagram shows 34 people. One person is selected at random. Find the following probability. Give your answer as a fraction in its simplest form:

$$P(A \cup B')$$

